

AHSAN RIAZ

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EDUCATION

National University of Sciences and Technology (NUST), SEECS

Islamabad, Pakistan

Bachelor of Computer Science (BSCS) | Expected May 2027 | CGPA: 3.23 / 4.00

Relevant coursework: Machine Learning, Deep Learning, Big Data Analytics, Distributed Systems, Compiler Construction, Software Engineering, Data Structures & Algorithms, Databases, Operating Systems, Networks

Punjab Group of Colleges

Faisalabad, Pakistan

Intermediate of Computer Science | Grade: A | 2023

RESEARCH INTERESTS

Applied machine learning and deep learning; large language models and retrieval-augmented generation (RAG); AI agent systems and data pipelines; scalable ML infrastructure.

EXPERIENCE

Software Developer Intern — Aawaz AI

Jun 2025 - Aug 2025

NSTP, NUST, Islamabad

- Built an end-to-end AI agent for business workflows (idea generation, landing pages, ad campaigns, outreach automation).
- Developed automation pipelines to extract structured data from bank statement PDFs (OCR and parsing).
- Integrated LLM APIs for multi-step task execution; supported cloud deployment and production.
- Built full-stack features with React, Node.js/Express, and databases for an AI-driven product.

Freelance ML & Automation Engineer — Upwork (Top Rated, 100% JSS) & Fiverr

Jun 2023 - Present

- Delivered 30+ projects across AI integrations, automation, web scraping, and full-stack apps, with a 100% Job Success Score.
- Built AI solutions with OpenAI, Claude, Gemini, LangChain, n8n, and Make.com for chatbots and workflow automation.
- Engineered Python pipelines for data extraction, processing, and database integration (PostgreSQL, MongoDB).

PROJECTS

Clutch.ai — Real-time Interview Co-pilot (Deep Learning)

- Built a real-time assistance system using a BiLSTM intent classifier, a triplet-trained MiniLM retriever, and a cross-attention reranker for context-aware response generation.

Tech: Python, PyTorch, Transformers, MiniLM

US Corn Yield Prediction — Machine Learning Research Project

- Integrated four public data sources (USDA NASS, NASA POWER, NRCS soil, Census TIGER) into 82,000+ county-year records spanning 1981 to 2023.
- Engineered climate-anomaly and crop-stress features; trained and tuned Ridge, Random Forest, Gradient Boosting, and XGBoost with a leakage-safe temporal split.
- Best model (XGBoost) reached $R^2 = 0.863$; deployed an interactive prediction app.

Tech: Python, scikit-learn, XGBoost, Streamlit

Giraph — AI Business Analytics Platform

- Full-stack analytics app transforming CSV/Excel data via Gemini AI; Node/Express + MongoDB backend auto-processes files, generates schemas, and recommends visualizations, with a Next.js/React frontend.

Tech: Next.js, React, TypeScript, Node.js, Express, MongoDB, Gemini AI, Firebase, ECharts

SehatYaad — Cross-Platform Medication Adherence App

- React Native/Expo app with prescription OCR, adaptive reminders, voice input, bilingual EN/UR support (TTS, RTL), adherence analytics, and a caregiver alert system; Flask backend.

Tech: React Native, Expo, TypeScript, PostgreSQL, Flask, Gemini AI

RevAutoSale — Large-Scale Scraping Backend

- Django backend with Scrapy/Selenium scrapers maintaining 100,000+ Copart vehicle listings; MySQL database and Django REST APIs for inventory and user functions.

Tech: Python, Django, Django REST, MySQL, Scrapy, Selenium

Depot — AI Infrastructure Architect (Final Year Project, in progress)

- A conversational system that takes developers from codebase to production-grade cloud infrastructure, aiming to do for infrastructure what code assistants did for writing code.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C, SQL, Bash

AI / ML: Deep learning (BiLSTM, Transformers), LLMs, RAG, vector embeddings, LangChain, OpenAI/Claude/Gemini APIs, OpenCV, Hugging Face, scikit-learn

Frontend: React, React Native, Next.js, Redux, Tailwind CSS, Radix UI, Recharts

Backend: Node.js, Express.js, Django, Flask, FastAPI, Socket.io

Data & Infra: PostgreSQL, MySQL, MongoDB, Redis, Firestore, Docker, AWS (EC2, S3, Lambda, IAM), NGINX, Git/GitHub, Linux

Scraping & Automation: Scrapy, Selenium, Playwright, BeautifulSoup, Celery, n8n, Make.com, Pandas, NumPy

LEADERSHIP & ACTIVITIES

Director — NUST Community Service Club (NCSC)

- Led the university's community service club, directing volunteer teams and organizing community initiatives.

HONORS & AWARDS

- 3rd Place, Vyrothon 2026 (ImagineArt / Vyro.ai).
- Top Rated freelancer on Upwork (100% Job Success Score, 30+ projects).
- Participant, Google Antigravity Hackathon.